Problem 7

Let $\mathcal{G}$ be a $2 \times 2 \times 2$ tensor with slices:

$$\mathcal{G}(:,:,1) = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \quad \mathcal{G}(:,:,2) = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}$$

Furthermore, let: $A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \quad B = \begin{bmatrix} -2 & 2 \\ 3 & -3 \\ -4 & -4 \end{bmatrix} \quad$ and $C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Calculate $\mathcal{G} \times_1 A \times_2 B \times_3 C$

Answer: $\mathcal{G} \times_1 A \in \mathbb{R}^{3\times 2\times 2}$.

$$(\mathcal{G} \times_1 A)_{111} = a_{11}\mathcal{G}_{111} + a_{12}\mathcal{G}_{211} = 1(1) + 4(-2) = -7$$

$$(\mathcal{G} \times_1 A)_{112} = a_{11}\mathcal{G}_{112} + a_{12}\mathcal{G}_{212} = 1(-2) + 4(1) = 2$$

$$(\mathcal{G} \times_1 A)_{121} = a_{11}\mathcal{G}_{121} + a_{12}\mathcal{G}_{221} = 1(2) + 4(1) = 6$$

$$(\mathcal{G} \times_1 A)_{122} = a_{11}\mathcal{G}_{122} + a_{12}\mathcal{G}_{222} = 1(1) + 4(2) = 9$$

$$(\mathcal{G} \times_1 A)_{211} = a_{21}\mathcal{G}_{111} + a_{22}\mathcal{G}_{211} = 2(1) + 5(-2) = -8$$

$$(\mathcal{G} \times_1 A)_{212} = a_{21}\mathcal{G}_{112} + a_{22}\mathcal{G}_{212} = 2(-2) + 5(1) = 1$$

$$(\mathcal{G} \times_1 A)_{221} = a_{21}\mathcal{G}_{121} + a_{22}\mathcal{G}_{221} = 2(2) + 5(1) = 9$$

$$(\mathcal{G} \times_1 A)_{222} = a_{21}\mathcal{G}_{122} + a_{22}\mathcal{G}_{222} = 2(1) + 5(2) = 12$$

$$(\mathcal{G} \times_1 A)_{311} = a_{31}\mathcal{G}_{111} + a_{32}\mathcal{G}_{211} = 3(1) + 6(-2) = -9$$

$$(\mathcal{G} \times_1 A)_{312} = a_{31}\mathcal{G}_{112} + a_{32}\mathcal{G}_{212} = 3(-2) + 6(1) = 0$$

$$(\mathcal{G} \times_1 A)_{321} = a_{31}\mathcal{G}_{121} + a_{32}\mathcal{G}_{221} = 3(2) + 6(1) = 12$$

$$(\mathcal{G} \times_1 A)_{322} = a_{31}\mathcal{G}_{122} + a_{32}\mathcal{G}_{222} = 3(1) + 6(2) = 15$$

Let $\mathcal{P} = (\mathcal{G} \times_1 A)$. Then $\mathcal{P}(:,:,1) = \begin{bmatrix} -7 & 6 \\ -8 & 9 \\ -9 & 12 \end{bmatrix}$ and $\mathcal{P}(:,:,2) = \begin{bmatrix} 2 & 9 \\ 1 & 12 \\ 0 & 15 \end{bmatrix}$

Next, we solve $\mathcal{P} \times_2 B \in \mathbb{R}^{3\times 3\times 2}$. Recall that $B = \begin{bmatrix} -2 & 2 \\ 3 & -3 \\ -4 & -4 \end{bmatrix}$

Also recall that $(\mathcal{P} \times_2 B)_{ijh} = \sum_{s=1}^{m} b_{js} p_{ish}$

$$(\mathcal{P} \times_2 B)_{111} = b_{11}p_{111} + b_{12}p_{121} = (-2)(-7) + (2)(6) = 26$$

$$(\mathcal{P} \times_2 B)_{112} = b_{11}p_{112} + b_{12}p_{122} = (-2)(2) + (2)(9) = 14$$

$$(\mathcal{P} \times_2 B)_{121} = b_{21}p_{111} + b_{22}p_{121} = (3)(-7) + (-3)(6) = -39$$

$$(\mathcal{P} \times_2 B)_{122} = b_{21}p_{112} + b_{22}p_{122} = (3)(2) + (-3)(9) = -21$$

$$(\mathcal{P} \times_2 B)_{131} = b_{31}p_{111} + b_{32}p_{121} = (-4)(-7) + (-4)(6) = 4$$

$$(\mathcal{P} \times_2 B)_{132} = b_{31}p_{112} + b_{32}p_{122} = (-4)(2) + (-4)(9) = -44$$

$$(\mathcal{P} \times_2 B)_{211} = b_{11}p_{211} + b_{12}p_{221} = (-2)(-8) + (2)(9) = 34$$

$$(\mathcal{P} \times_2 B)_{212} = b_{11}p_{212} + b_{12}p_{222} = (-2)(1) + (2)(12) = 22$$

$$(\mathcal{P} \times_2 B)_{221} = b_{21}p_{211} + b_{22}p_{221} = (3)(-8) + (-3)(9) = -51$$

$$(\mathcal{P} \times_2 B)_{222} = b_{21}p_{212} + b_{22}p_{222} = (3)(1) + (-3)(12) = -33$$

$$(\mathcal{P} \times_2 B)_{231} = b_{31}p_{211} + b_{32}p_{221} = (-4)(-8) + (-4)(9) = -4$$

$$(\mathcal{P} \times_2 B)_{232} = b_{31}p_{212} + b_{32}p_{222} = (-4)(1) + (-4)(12) = -52$$

$$(\mathcal{P} \times_2 B)_{311} = b_{11}p_{311} + b_{12}p_{321} = (-2)(-9) + (2)(12) = 42$$

$$(\mathcal{P} \times_2 B)_{312} = b_{11}p_{312} + b_{12}p_{322} = (-2)(0) + (2)(15) = 30$$

$$(\mathcal{P} \times_2 B)_{321} = b_{21}p_{311} + b_{22}p_{321} = (3)(-9) + (-3)(12) = -63$$

$$(\mathcal{P} \times_2 B)_{322} = b_{21}p_{312} + b_{22}p_{322} = (3)(0) + (-3)(15) = -45$$

$$(\mathcal{P} \times_2 B)_{331} = b_{31}p_{311} + b_{32}p_{321} = (-4)(-9) + (-4)(12) = -12$$

$$(\mathcal{P} \times_2 B)_{332} = b_{31}p_{312} + b_{32}p_{322} = (-4)(0) + (-4)(15) = -60$$

Let $\mathcal{Q} = (\mathcal{P} \times_2 B)$. Then $\mathcal{Q}(:,:,1) = \begin{bmatrix} 26 & -39 & 4 \\ 34 & -51 & -4 \\ 42 & -63 & -12 \end{bmatrix}$ and $\mathcal{Q}(:,:,2) = \begin{bmatrix} 14 & -21 & -44 \\ 22 & -33 & -52 \\ 30 & -45 & -60 \end{bmatrix}$

Finally, we must compute $\mathcal{Q} \times_3 C \in \mathbb{R}^{3\times3\times2}$. Recall that $C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

Also recall that $(\mathcal{Q} \times_3 C)_{ijh} = \sum_{s=1}^{m} c_{hs} q_{ijs}$.

$(\mathcal{Q} \times_3 C)_{111} = c_{11}q_{111} + c_{12}q_{112} = (2)(26) + (0)(14) = 52$

$(\mathcal{Q} \times_3 C)_{112} = c_{21}q_{111} + c_{22}q_{112} = (0)(26) + (-1)(14) = -14$

$(\mathcal{Q} \times_3 C)_{121} = c_{11}q_{121} + c_{12}q_{122} = (2)(-39) + (0)(-21) = -78$

$(\mathcal{Q} \times_3 C)_{122} = c_{21}q_{121} + c_{22}q_{122} = (0)(-39) + (-1)(-21) = 21$

$(\mathcal{Q} \times_3 C)_{131} = c_{11}q_{131} + c_{12}q_{132} = (2)(4) + (0)(-44) = 8$

$(\mathcal{Q} \times_3 C)_{132} = c_{21}q_{131} + c_{22}q_{132} = (0)(4) + (-1)(-44) = 44$

$(\mathcal{Q} \times_3 C)_{211} = c_{11}q_{211} + c_{12}q_{212} = (2)(34) + (0)(22) = 68$

$(\mathcal{Q} \times_3 C)_{212} = c_{21}q_{211} + c_{22}q_{212} = (0)(34) + (-1)(22) = -22$

$(\mathcal{Q} \times_3 C)_{221} = c_{11}q_{221} + c_{12}q_{222} = (2)(-51) + (0)(-33) = -102$

$(\mathcal{Q} \times_3 C)_{222} = c_{21}q_{221} + c_{22}q_{222} = (0)(-51) + (-1)(-33) = 33$

$(\mathcal{Q} \times_3 C)_{231} = c_{11}q_{231} + c_{12}q_{232} = (2)(-4) + (0)(-52) = -8$

$(\mathcal{Q} \times_3 C)_{232} = c_{21}q_{231} + c_{22}q_{232} = (0)(-4) + (-1)(-52) = 52$

$(\mathcal{Q} \times_3 C)_{311} = c_{11}q_{311} + c_{12}q_{312} = (2)(42) + (0)(30) = 84$

$(\mathcal{Q} \times_3 C)_{312} = c_{21}q_{311} + c_{22}q_{312} = (0)(42) + (-1)(30) = -30$

$(\mathcal{Q} \times_3 C)_{321} = c_{11}q_{321} + c_{12}q_{322} = (2)(-63) + (0)(-45) = -126$

$(\mathcal{Q} \times_3 C)_{322} = c_{21}q_{331} + c_{22}q_{332} = (0)(-63) + (-1)(-45) = 45$

$(\mathcal{Q} \times_3 C)_{331} = c_{11}q_{331} + c_{12}q_{332} = (2)(-12) + (0)(-60) = -24$

$(\mathcal{Q} \times_3 C)_{332} = c_{21}q_{331} + c_{22}q_{332} = (0)(-12) + (-1)(-60) = 60$

Let $\mathcal{R} = (\mathcal{Q} \times_3 C)$. Then $\mathcal{R}(:,:,1) = \begin{bmatrix} 52 & -78 & 8 \\ 68 & -102 & -8 \\ 84 & -126 & -24 \end{bmatrix}$ and $\mathcal{R}(:,:,2) = \begin{bmatrix} -14 & 21 & 44 \\ -22 & 33 & 52 \\ -30 & 45 & 60 \end{bmatrix}$

Finally, we can conclude that $\mathcal{G} \times_1 A \times_2 B \times_3 C = \mathcal{R}$ which is defined above.